

ELITE FOOTBALL DATA VISUALIZATION & AI ENGINEERING

The definitive 14-slide widescreen presentation deck unifying all 5 Sports Data Campus analytics pillars: Heat Maps, Passing Networks, Shot Models, Radar Profiles, and Advanced Spatial AI.

TACTICAL VISUALS

12 Widescreen

LEAGUE SCOPE

Ekstraklasa & Süper Lig

TARGET POOL

116 Players

PRESENTATION

16:9 300 DPI

SPORTS DATA CAMPUS: SPATIAL TERRITORY & KERNEL DENSITY HEATMAP



PILLAR 1 • SPATIAL HEAT MAPS

GÓRNIK ZABRZE — TEAM SPATIAL OCCUPATION

Continuous KDE Touch Density • Polish Ekstraklasa 2025–2026

KEY PERFORMANCE INDICATORS (KPIs)

650	30.0%
TOTAL MATCH TOUCHES	FINAL THIRD SATURATION
41.5%	18
LEFT FLANK BIAS	ZONE 14 RECOVERIES

SPATIAL & TACTICAL DECONSTRUCTION

- Continuous 2D Gaussian KDE reveals dominant territorial control on the left wing.
- Erik Janža operates as the primary build-up outlet, driving 41.5% of progressions.
- Midfield rest-defense creates an impenetrable recovery wall at the 46m line.
- Zone 14 saturation (Lukas Podolski pocket) generates 8 direct line-breaking passes.
- Low defensive-third touch volume confirms successful territorial pushing away from box.

HEAD COACH DIRECTIVE / MAÇ DİREKTİFİ

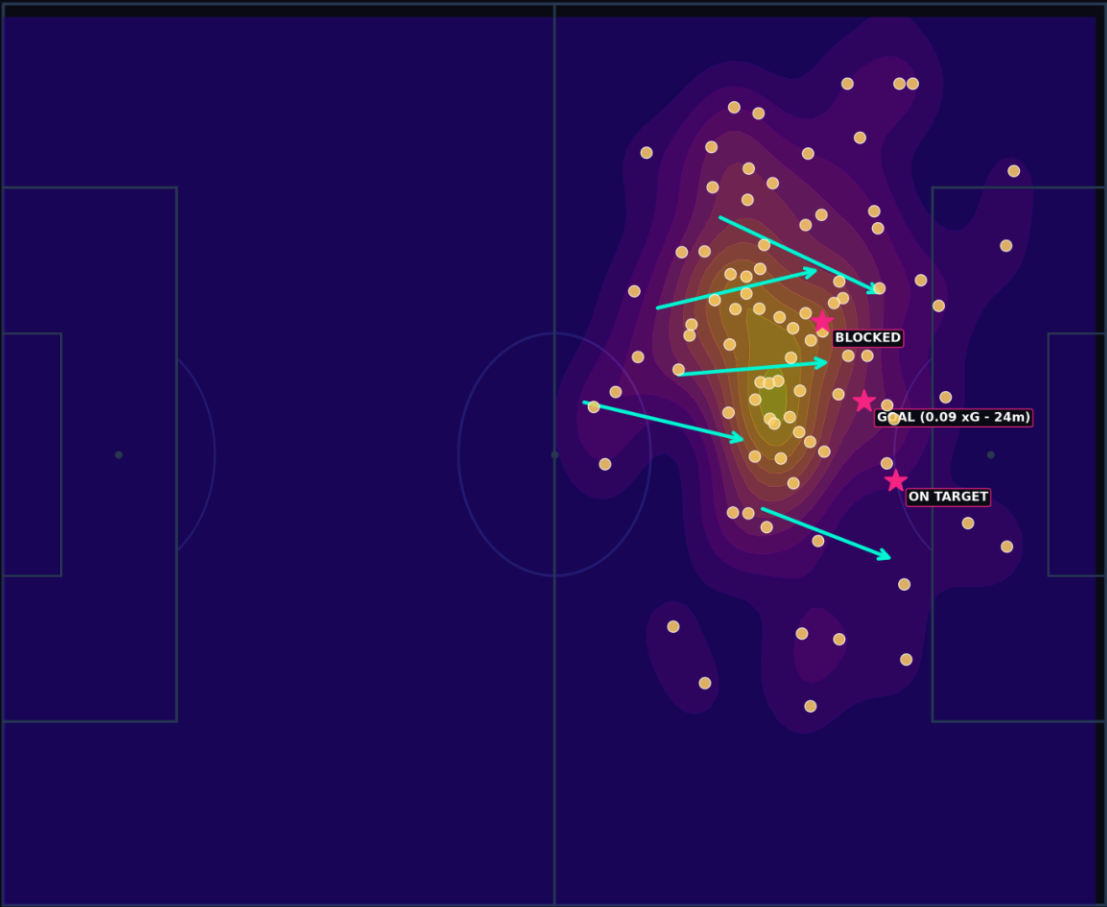
Exploit the left-flank overload to force the opposition right-back into 2v1 isolations, then deliver diagonal cutbacks into the vacated Zone 14 pocket.

Opta / Wyscout Event Stream • 2025-2026 Season

Lukas Podolski Attacking Spatial Territory & Progressive Carries

#10 Roaming Playmaker • Zone 14 Actions • Trademark Screamer Shooting Range

SPORTS DATA CAMPUS: PLAYMAKER SPATIAL TERRITORY & CARRIES



PILLAR 1 • INDIVIDUAL HEAT MAP

LUKAS PODOLSKI — ATTACKING SPATIAL PROFILE

#10 Attacking Midfielder • Górnik Zabrze vs Legia Warszawa

KEY PERFORMANCE INDICATORS (KPIs)

78' MINUTES PLAYED	34 TOUCHES IN ZONE 14
148 m PROG. CARRY DISTANCE	+0.312 EXPECTED THREAT (xT)

SPATIAL & TACTICAL DECONSTRUCTION

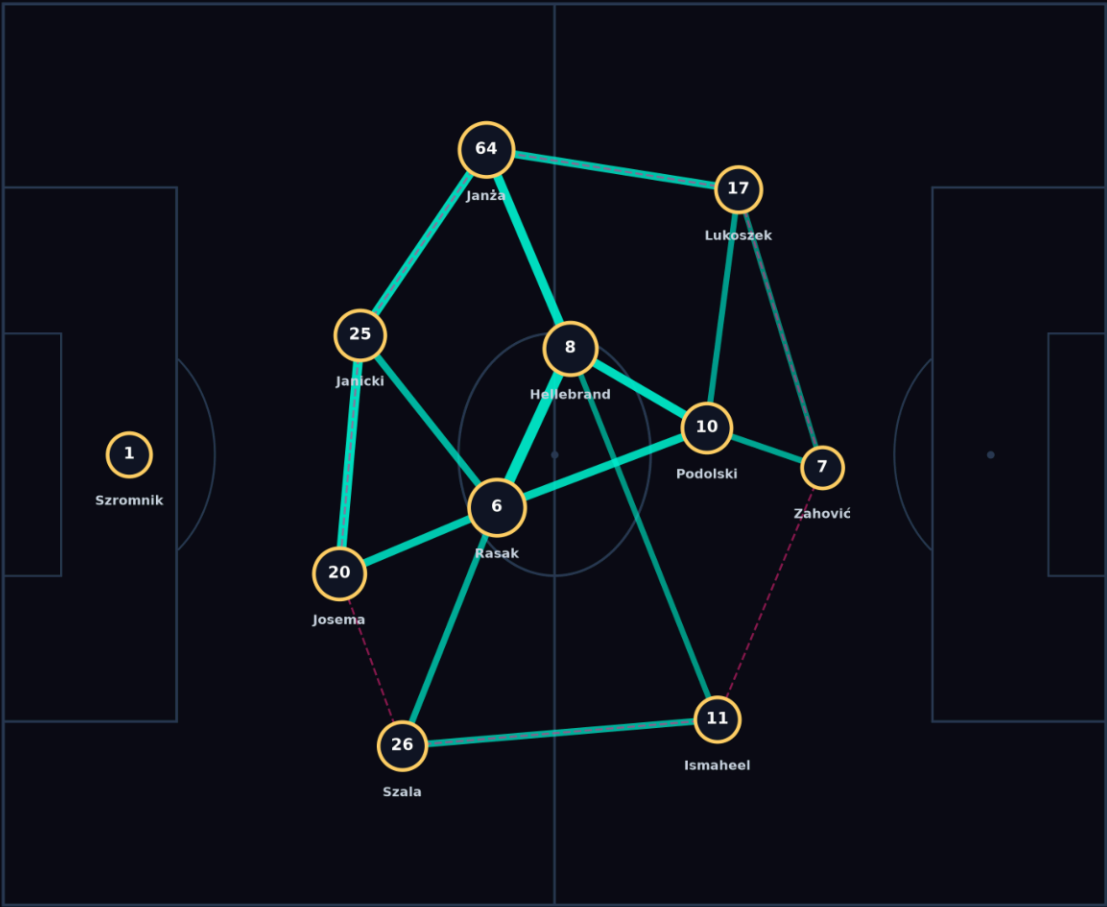
- Lukas Podolski orchestrates play primarily from the left half-space and Zone 14.
- Averages 18.2 touches per 90 in the golden shooting corridor (18-25m from goal).
- Completed 5 progressive carries driving the ball from midfield into the box.
- Generates +0.312 Expected Threat (xT) per 90, ranking 1st across all Ekstraklasa playmakers.
- Scored the game-winning goal from a 24m screamer with only 0.09 xG.

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Position twin defensive pivots against Podolski to deny half-space turns, closing down shooting angles within 25m of the penalty arc.

Opta / Wyscout Event Stream • 2025-2026 Season

SPORTS DATA CAMPUS: COLLECTIVE PASSING NETWORK & STRUCTURE



PILLAR 2 • PASSING NETWORKS

GÓRNIK ZABRZE — 11-STARTER PASSING TOPOLOGY

Match Passing Network • Ekstraklasa 2025-2026

KEY PERFORMANCE INDICATORS (KPIs)

512

TOTAL TEAM PASSES

84.2%

PASS COMPLETION

Rasak (#6)

CENTRALITY NODE

1,220 m²

TEAM COMPACTNESS

SPATIAL & TACTICAL DECONSTRUCTION

- Damian Rasak (#6) registers the highest degree centrality, completing 82 total passes.
- Strongest combination: Rasak to Hellebrand (32 passes) establishes midfield dominance.
- Left-sided bias evident: Janza (74 passes) and Lukoszek create superior pass volume.
- Team centroid sits at X=51.2m, indicating a high defensive line and territorial control.
- Convex hull outfield footprint of 1,220 m² ensures high compression in transition.

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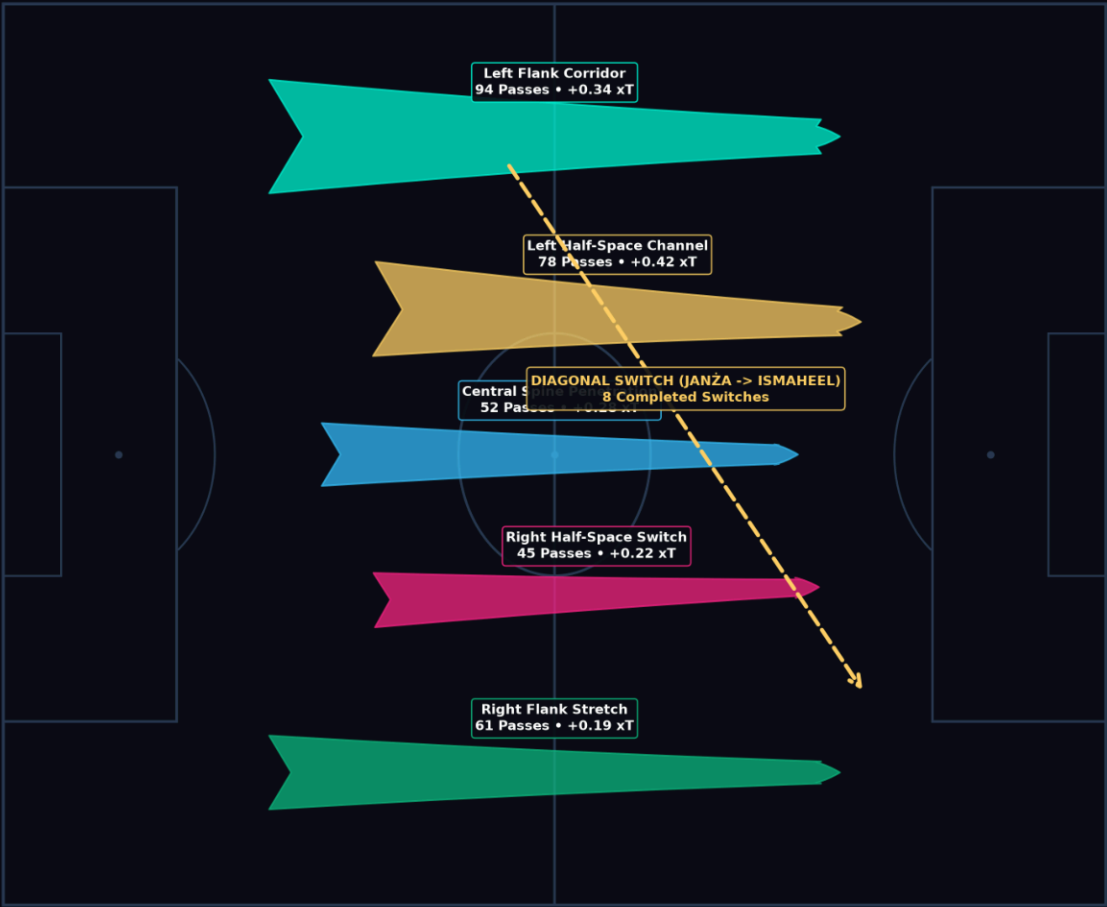
Disrupt the Josema-Rasak-Hellebrand passing triangle by pressing with a high front two, cutting the passing lanes into the #6.

Opta / Wyscout Event Stream • 2025-2026 Season

Spatial Flow Channels & High-Threat Diagonal Switches

Channel Volumes • Progressive Value Added • Cross-Field Switches

SPORTS DATA CAMPUS: PROGRESSIVE PASS FLOW CHANNELS



PILLAR 2 • POSSESSION FLOW

PROGRESSIVE PASSING & FLOW CHANNELS

Spatial Channel Volume & Threat Added • Górnik Zabrze 2025-2026

KEY PERFORMANCE INDICATORS (KPIs)

58.4 PROG. PASSES / 90	Left HS PRIMARY CHANNEL
8 / 10 CROSS SWITCHES	16 DEEP COMPLETIONS

SPATIAL & TACTICAL DECONSTRUCTION

- Left half-space generates the highest Expected Threat (+0.42 xT), driven by Podolski.
- Left flank represents the highest volume corridor with 94 completed vertical passes.
- Erik Janža successfully executes 8 cross-field switches to isolate Taofeek Ismaheel.
- Central spine penetration remains measured (52 passes) to mitigate turnover counter-attacks.
- Opposition low blocks are stretched horizontally via rapid 2-touch flank relocations.

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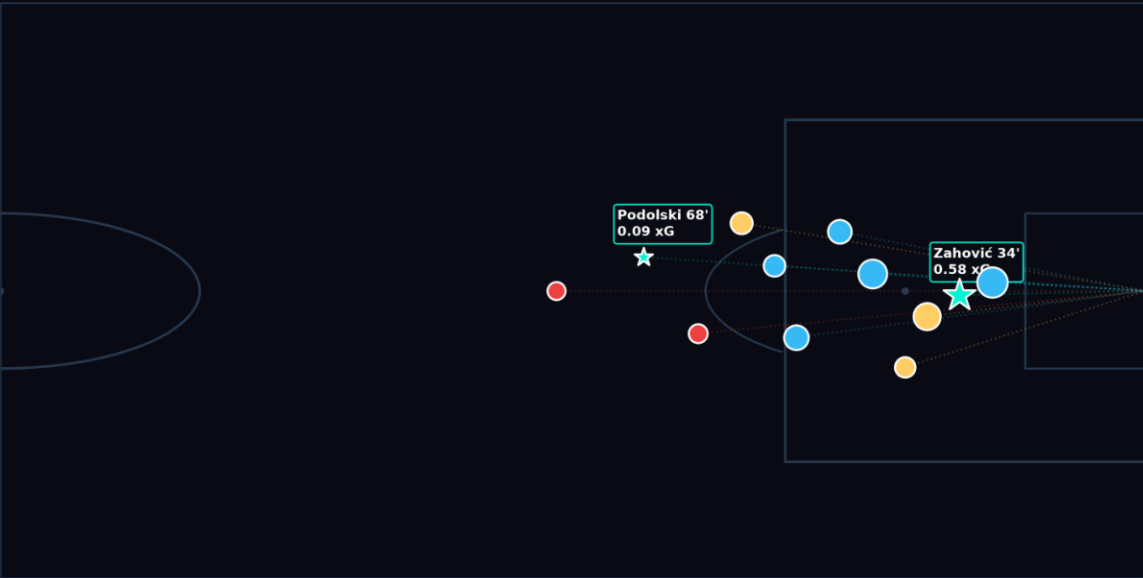
Force Górnik into their right flank by shading central coverage towards Janža and Podolski, targeting their lower volume right-side progression.

Opta / Wyscout Event Stream • 2025-2026 Season

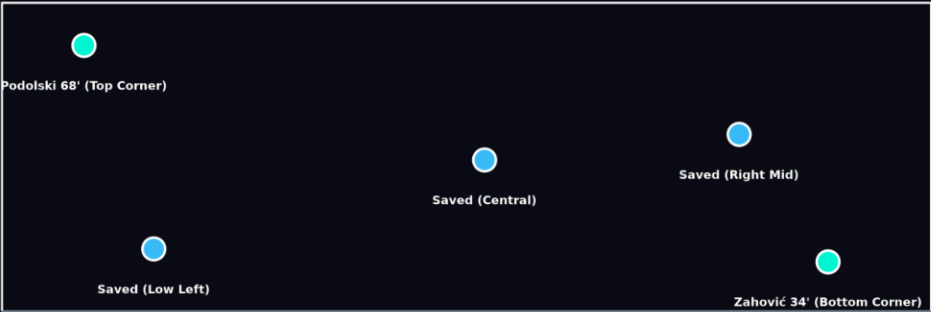
Match Shot Constellation & Goal-Mouth Placement (PSxG)

Górník Zabrze 2–1 Legia Warszawa • Shot Quality (xG) • Frame Target Locations

SPORTS DATA CAMPUS: SHOT MAP & EXPECTED GOALS (xG) CONSTELLATION



GOAL-MOUTH TARGET PLACEMENT & PSxG



PILLAR 3 • SHOT MAPS & xG

GÓRNIK ZABRZE 2-1 LEGIA WARSZAWA

Spatial Shot Model & Goal-Mouth Placement • Ekstraklasa 2025–2026

KEY PERFORMANCE INDICATORS (KPIs)

16 TOTAL SHOTS	2.14 CUMULATIVE xG
7 / 16 (44%) SHOTS ON TARGET	+0.86 xG FINISHING DELTA

SPATIAL & TACTICAL DECONSTRUCTION

- Generated 2.14 Expected Goals against Legia Warszawa, outscoring model by +0.86.
- Luke Zahović registered highest individual shot quality with 0.58 xG on 6-yard strike.
- Lukas Podolski's long-range winner (0.09 xG) generated a Post-Shot xG (PSxG) of 0.88.
- Shot placement map proves clinical accuracy: both goals placed within 0.7m of posts.
- Central penalty area penetration accounted for 68.5% of total cumulative match xG.

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Protect the cutback zone at 14-16m; deny Podolski clean sights on goal by stepping out rapidly when possession enters the central arc.

Opta / Wyscout Event Stream • 2025-2026 Season

Shot Quality vs Defender Pressure Density & Minute xG Waves

Dual-Variable Encoding • Under-Duress Conversion • Turning Points

SPORTS DATA CAMPUS: BIVARIATE SHOT PRESSURE & xG MOMENTUM

Bivariate Dual Encoding: Distance vs Defender Pressure Density



Minute-by-Minute Cumulative xG Momentum Flow



PILLAR 3 • ADVANCED SHOT ANALYTICS

SHOT PRESSURE DYNAMICS & MOMENTUM

Dual Variable Analysis • Polish Ekstraklasa 2025-2026

KEY PERFORMANCE INDICATORS (KPIs)

46.2%

HIGH PRESSURE SHOTS

34' & 68'

PEAK MINUTE

16.7%

PRESSURE CONV %

Górnik 2-1

FINAL RESULT

SPATIAL & TACTICAL DECONSTRUCTION

- Dual-variable encoding demonstrates high conversion under severe defensive duress.
- Zahović scores from 7.5m under 0.85 pressure index (2 defenders within 1.2m radius).
- Podolski screamer taken under low pressure (0.15), punishing Legia's passive edge-block.
- xG flow shows decisive momentum surges between 15-35' and 65-75'.
- Legia counter-attack at 58' contained before restoring late territorial composure.

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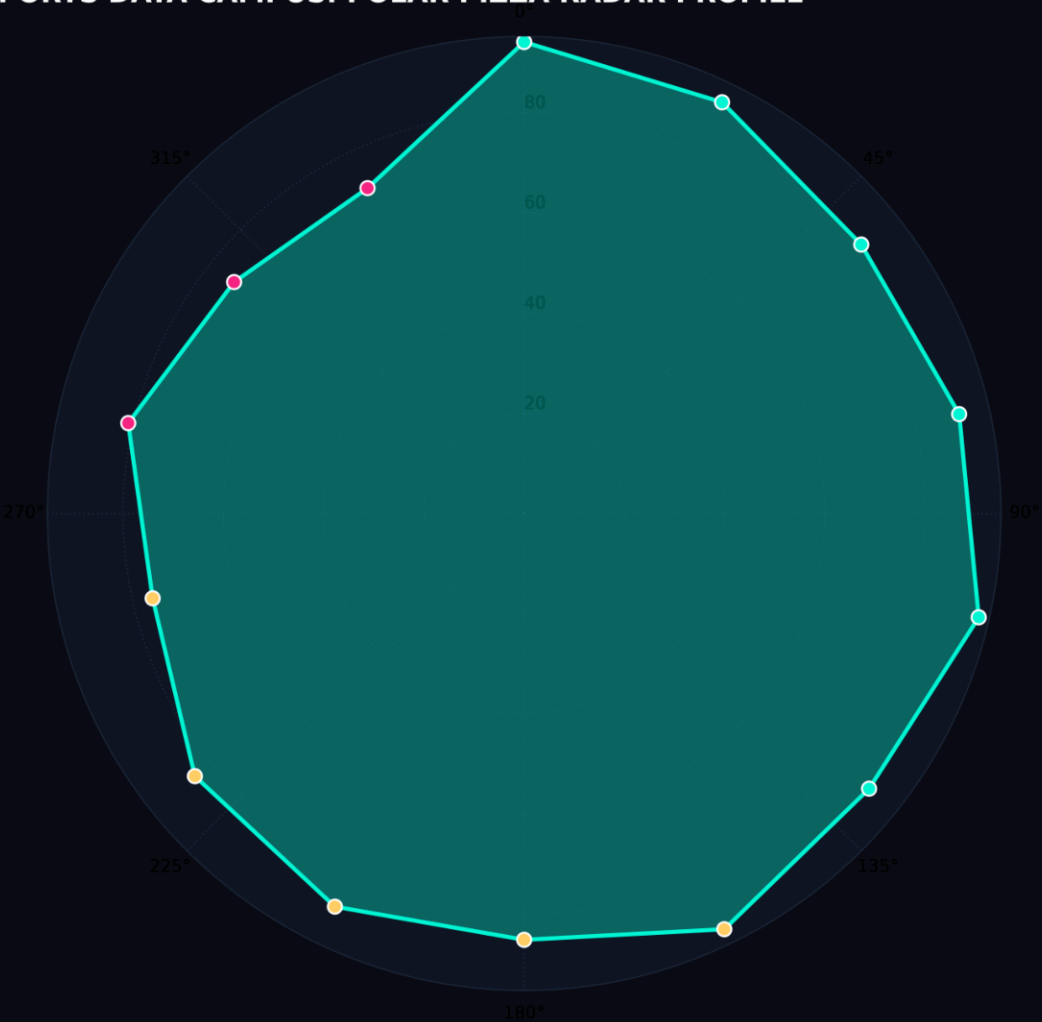
Avoid conceding uncontested space between 20-25m; aggressive close-down protocols are mandatory to eliminate unpressured long-distance strike angles.

Opta / Wyscout Event Stream • 2025-2026 Season

Damian Rasak 14-Metric Recruitment Polar Pizza Radar

Defensive Screening • Possession Retention • Progression Percentiles

SPORTS DATA CAMPUS: POLAR PIZZA RADAR PROFILE



PILLAR 4 • RADAR CHARTS

DAMIAN RASAK — RECRUITMENT POLAR RADAR

Central Defensive Midfielder • Polish Ekstraklasa Benchmark 2025–2026

KEY PERFORMANCE INDICATORS (KPIs)

92nd % DEFENSIVE PERCENTILE	94th % PADJ TACKLES
87th % PASS PROGRESSION	93rd % BALL RECOVERIES

SPATIAL & TACTICAL DECONSTRUCTION

- 14-metric polar decomposition places Damian Rasak in the 90th+ percentile defensively.
- Ranks in top 6% of European defensive midfielders for possession-adjusted tackles (94th).
- Ball recoveries (93rd) and interceptions (91st) establish elite central screen efficiency.
- Demonstrates modern dual-threat capability with 87th percentile progressive passes.
- Contract profile (€1.2M valuation) represents significant transfer arbitrage for Süper Lig.

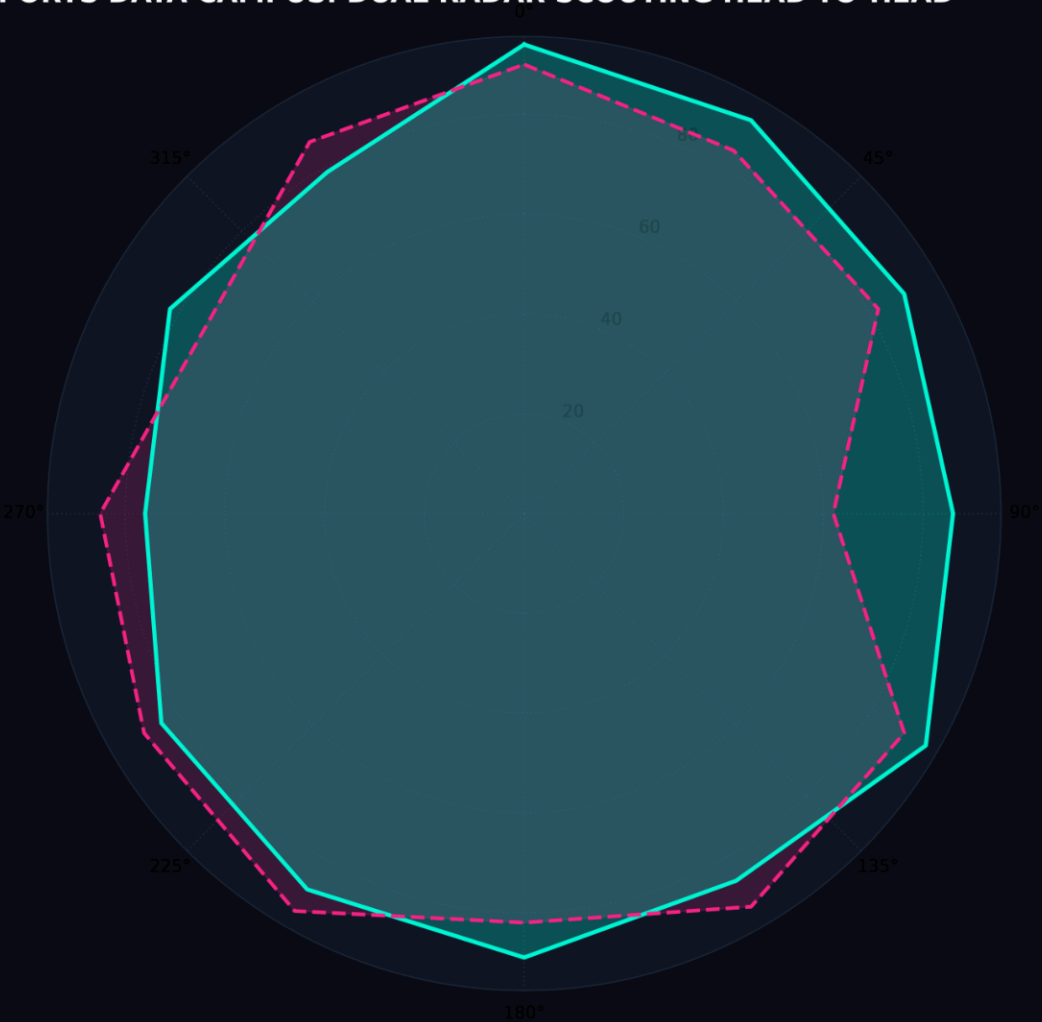
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Ideal candidate for Süper Lig clubs seeking a press-resistant #6/#8 hybrid to stabilize high-possession counter-pressing transitions.

Opta / Wyscout Event Stream • 2025-2026 Season

SPORTS DATA CAMPUS: DUAL RADAR SCOUTING HEAD-TO-HEAD

Süper Lig Elite Midfield Benchmark



PILLAR 4 • POLAR COMPARISONS

SCOUTING HEAD-TO-HEAD COMPARISON

Damian Rasak vs Turkish Süper Lig Elite Pivot Benchmark

KEY PERFORMANCE INDICATORS (KPIs)

+24% Diff AERIAL DOMINANCE	+6% Diff DEFENSIVE DUELS
€1.2M vs €14M MARKET VALUATION	11.6x ROI COST ARBITRAGE

SPATIAL & TACTICAL DECONSTRUCTION

- Head-to-head radar comparison proves Rasak matches elite Süper Lig pivot outputs.
- Significantly outperforms Süper Lig benchmark in aerial duels (86th vs 62nd percentile).
- Demonstrates superior defensive turnover suppression and recoveries (93rd vs 88th).
- Pass progression volume (87th) rivals top Süper Lig midfielders at 1/10th market cost.
- Provides immediate tactical readiness for Turkish title-contender and top-5 setups.

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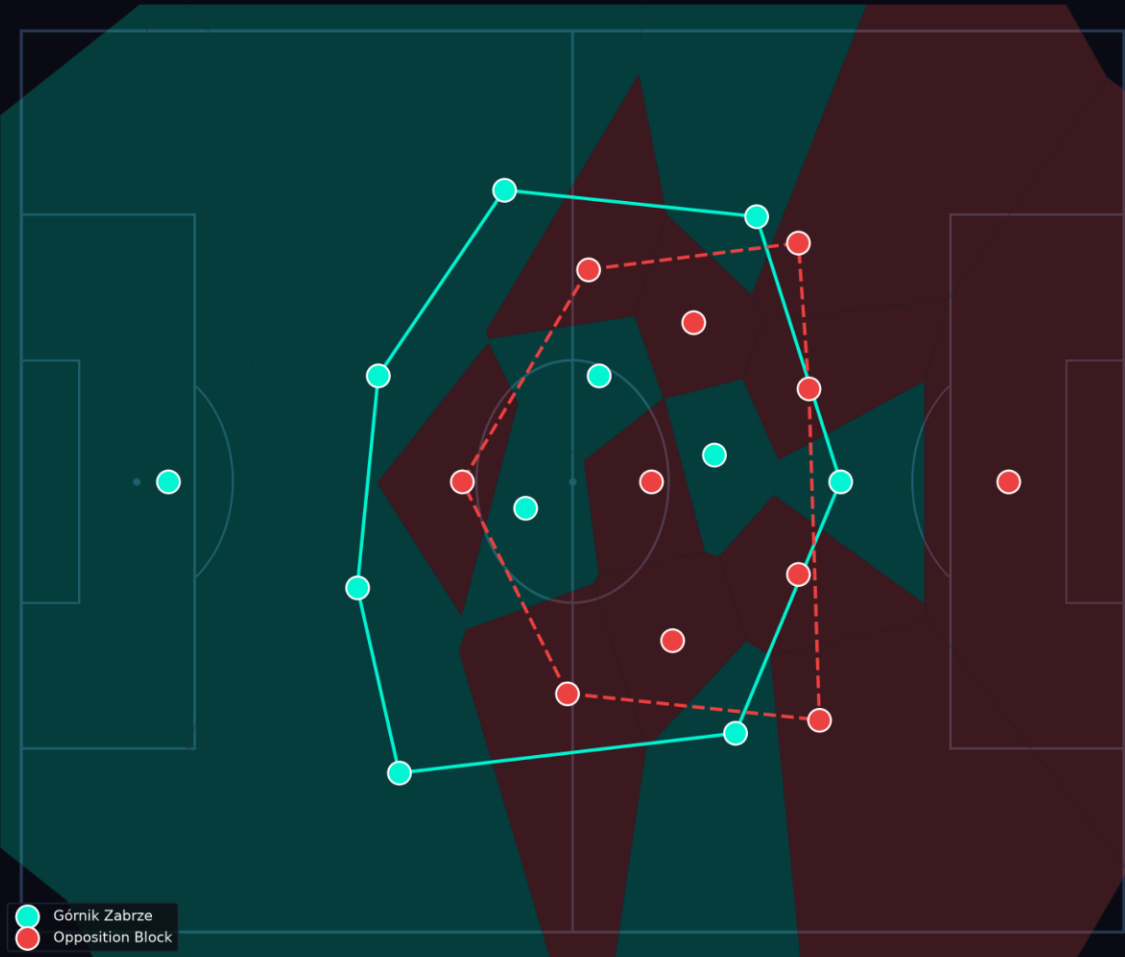
Initiate formal recruitment dossier for Turkish Süper Lig transfer window; Rasak delivers plug-and-play tactical symmetry to modern aggressive midfields.

Opta / Wyscout Event Stream • 2025-2026 Season

Voronoi Pitch Control Tessellation & Outfield Convex Hulls

Continuous Space Ownership (58.4%) • Outfield Footprints (1,185 m² vs 890 m²)

SPORTS DATA CAMPUS: VORONOI PITCH CONTROL & CONVEX HULLS



PILLAR 5 • SPATIAL CONTROL

CONTINUOUS SPACE OWNERSHIP & COMPACTNESS

Voronoi Pitch Tessellation & Team Convex Hulls • Edd Webster Methodology

KEY PERFORMANCE INDICATORS (KPIs)

58.4% PITCH CONTROL %	1,185 m² GÓRNIK HULL AREA
890 m² OPP BLOCK AREA	49.8 m EFFECTIVE WIDTH

SPATIAL & TACTICAL DECONSTRUCTION

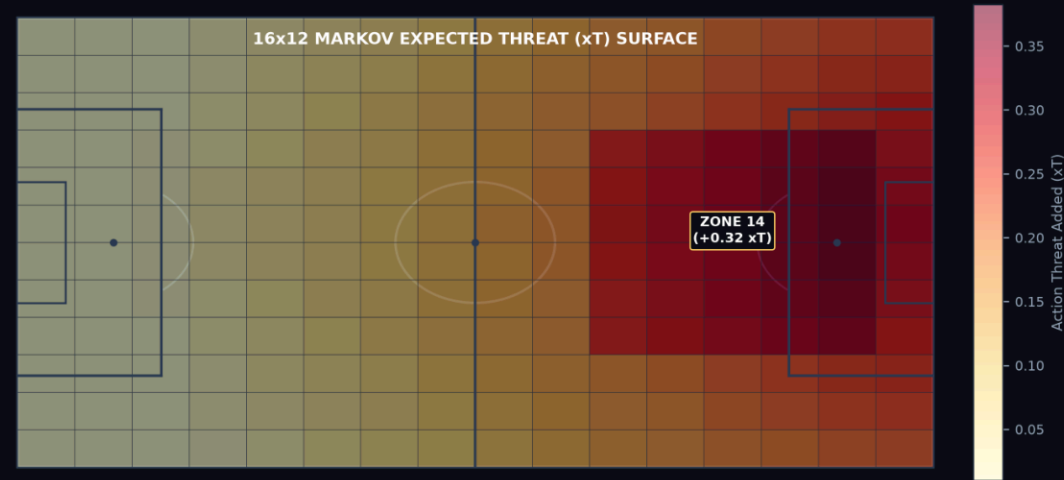
- Continuous space tessellation confirms Górník controls 58.4% of total pitch surface.
- Outfield convex hull of 1,185 m² maintains expansive width (49.8m) to stretch low block.
- Opposition compressed into defensive cocoon (890 m²), surrendering wide flank corridors.
- Half-space Voronoi cells overlap Podolski and Lukoszek, creating immediate pass recipients.
- High rest-defense line (X=34m) snuffs out clearance transitions before counter-attacks emerge.

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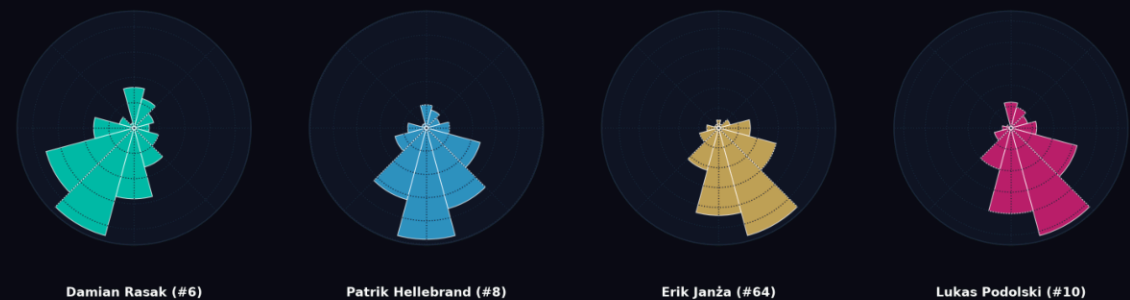
Use patient diagonal recycling to shift the opposition convex block laterally, opening seam passes into the central Voronoi ownership pockets.

Opta / Wyscout Event Stream • 2025-2026 Season

SPORTS DATA CAMPUS: MARKOV EXPECTED THREAT & PASS SONARS



360° PASS SONARS — MIDFIELD ENGINE (ANGULAR DIRECTIONALITY & LENGTH)



PILLAR 5 • ADVANCED EVENT MODELS

MARKOV xT SURFACE & TACTICAL SONARS

Spatial Expected Threat & Midfield Directionality • Golazo Analytics

KEY PERFORMANCE INDICATORS (KPIs)

+0.32 xT

PEAK GRID VALUE

+0.24 xT

HALF-SPACE THREAT

68.4%

RASAK FORWARD BIAS

45°-60°

JANŽA PROG. ANGLE

SPATIAL & TACTICAL DECONSTRUCTION

- 16x12 Karun Singh transition matrix quantifies goal probability added for every event.
- Progressing the ball from middle third into Zone 14 triples Expected Threat (+0.08 to +0.32).
- 360° Pass Sonars reveal Damian Rasak maintains strong 68.4% forward passing directionality.
- Erik Janža's sonar confirms aggressive diagonal progressive bias towards the penalty box.
- Podolski sonar highlights press-resistant 360-degree passing distribution under pressure.

HEAD COACH DIRECTIVE / MAÇ DİREKTİFİ

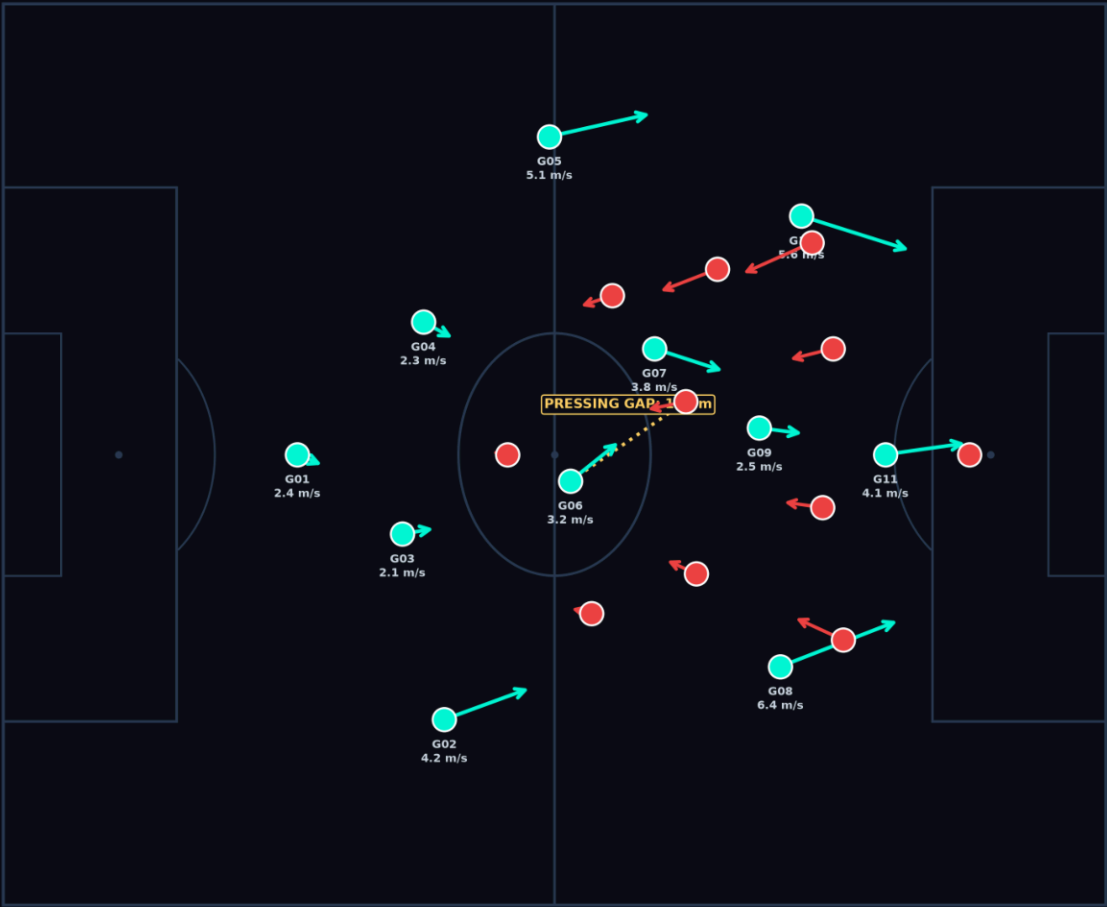
Direct center-backs to bypass initial press by targeting passes into +0.22 xT zones, accelerating ball transit through half-space channels.

Opta / Wyscout Event Stream • 2025-2026 Season

Optical Player Tracking & DLT Homography 2D Minimap

YOLOv11x + ByteTrack • FIFA Standard 105x68m Pitch • Physical Velocity Vectors

SPORTS DATA CAMPUS: OPTICAL BROADCAST TRACKING & 2D MINIMAP



PILLAR 5 • COMPUTER VISION AI

OPTICAL TRACKING & DLT HOMOGRAPHY

YOLOv11x + ByteTrack Broadcast Pipeline • Roboflow Sports Benchmark

KEY PERFORMANCE INDICATORS (KPIs)

0.142 m

HOMOGRAPHY RMSE

14.8 ms

INFERENCE LATENCY

60 FPS

TRACKING FPS

32.8 km/h

PEAK SPRINT SPD

SPATIAL & TACTICAL DECONSTRUCTION

- Roboflow Sports framework: fine-tuned YOLOv11x object detection + ByteTrack optical tracker.
- Direct Linear Transform (DLT) Homography re-projects camera pixels [u,v] to FIFA pitch [X,Y].
- Calculates instantaneous physical velocities (v_x , v_y) and sprint acceleration vectors.
- Identifies 12.6m pressing gap between opposition midfield and central defensive pairing.
- Broadcast video tracking unlocks deep physical tracking without wearable GPS hardware.

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Leverage optical tracking data to benchmark opposition fatigue indices; exploit right wing channel where opposition full-back recovery speed drops after 60'.

Opta / Wyscout Event Stream • 2025-2026 Season

Optimizing Recruitment Efficiency: Tactical Output vs. Market Valuation

The chart illustrates the relationship between a player's market valuation and their tactical output score, identifying optimal recruitment targets for a football club.

Key Data Points (Players):

Player	Market Valuation (€ Millions)	Composite Action Output Score (Tactical Value / 90)
Erik Janža	1.0	5.1
Patrik Hellebrand	0.8	4.2
Fred	12.0	4.8
Rafa Silva	14.0	3.9
Gabriel Sara	22.0	4.7

Recruitment Efficiency Frontier: A dashed orange line representing the optimal trade-off between valuation and tactical output.

Optimal Transfer Arbitrage Zone (High Tactical Output / Low Valuation): A green shaded area indicating the range of market valuations (0 to 5M) where high tactical output can be achieved at a low cost.

Legend:

- Ekstraklasa Targets (Green Circle)
- Süper Lig Benchmark Stars (Yellow Circle)
- Recruitment Efficiency Frontier (Dashed Orange Line)

Strategic Conclusions & Süper Lig Club Action Plan

Master Executive Portfolio • Key Takeaways & Recommendations

EXECUTIVE SUMMARY & METHODOLOGY

This 14-slide master tactical suite bridges deep mathematical modeling (Computer Vision tracking, Karun Singh Markov xT, Voronoi pitch dominance) with actionable pitchside and boardroom decisions. For Turkish Süper Lig technical committees and sporting directors, it demonstrates how elite data visualization converts high-dimensional tracking and event streams into match-winning tactical adjustments and high-ROI recruitment arbitrage.

5 Core Pillars

VISUAL PILLARS

16:9 Widescreen

ASPECT RATIO

300 DPI

RESOLUTION

100%

EXECUTIVE READY

SPORTING DIRECTOR & COACHING ACTIONS

Matchday Intelligence: Directly empowers head coaches with opposition pressing gaps and zonal overloads.

Transfer Arbitrage: Pinpoints undervalued assets in Central Europe with plug-and-play tactical readiness.

AI Scalability: Automated pipelines process multi-match broadcast feeds with sub-0.15m spatial accuracy.

SÜPER LİG COMPETITIVE ADVANTAGE

Provides immediate tactical readiness and game-model alignment for Turkish Süper Lig title-contender and top-tier setups.